Bowen Wu

PhD Candidate in Computer Science · ETH Zürich

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Education

ETH Zürich 2023 – Expected Winter 2027
PhD Candidate in Computer Science *Zürich, Switzerland*

- Supervised by Prof. Gustavo Alonso and Prof. Ana Klimovic.
- Build analytical and vector databases for heterogeneous computing platforms, including CPUs, GPUs, and advanced networking (e.g., InfiniBand, RoCE). Co-design database systems and hardware accelerators.
- Explore the intersection of databases and AI workloads (e.g., semantic queries, text-to-SQL).

ETH Zürich 2020 – 2023
MSc in Computer Science with Distinction (GPA: 5.78/6) *Zürich, Switzerland*

The Chinese University of Hong Kong 2016 – 2020
BSc in Computer Science, First Class Honours (Major GPA: 3.85/4) *Hong Kong, China*

Experience

NVIDIA Corporation May 2026 – Aug 2026
Research Intern *Santa Clara, USA*

- Worked with the Architecture Research Group, led by Christos Kozyrakis.
- Characterized and optimized distributed data analytics on scale-up GPU systems, e.g., NVL72.
- Studied database–hardware co-design, including computation units, memory hierarchy, and interconnect.

Meta Platforms, Inc. May 2025 – Aug 2025
Software Engineer Intern *Menlo Park, USA*

- Worked on Velox, a high-performance, open-source query execution engine.
- Prototyped a GPU-based reader for the Nimble file format, a replacement for Apache Parquet and ORC. The reader supports key features such as decoding nested-encoded data, filter push-down, and data chunking.

Microsoft Gray Systems Lab May 2024 – Aug 2024
Research Intern *Redmond, USA*

- Researched and optimized distributed multi-GPU SQL query acceleration.
- Profiled and modeled heterogeneous interconnects (e.g., NVLink, PCIe, Ethernet) in GPU clusters to assess their impact on GPU-accelerated query execution.
- Proposed a novel approach to optimize data exchange over heterogeneous interconnects, significantly improving query execution efficiency in GPU clusters. A patent application has been filed for this technique.

Cablex AG (Swisscom) Aug 2022 – Oct 2022
Database Analyst (Part-time) *Zürich, Switzerland*

- Designed a unified relational data model to enable comprehensive data analysis across multiple business sections.

Amazon Web Services Sep 2021 – Feb 2022
Software Engineer Intern *Berlin, Germany*

- Optimized the query rewriter of Amazon Redshift, the petabyte-scale cloud data warehouse in AWS.
- Proposed a collection of rewriting algorithms for nested SQL queries and implemented them in Redshift using both C++ and a proprietary domain-specific language.
- Significantly improved performance on the terabyte-scale TPC-DS benchmark with the proposed algorithms.

Publications

Conference and Workshop Papers

CODAs: Computational Data Paths for Lakehouses *CIDR 2027 (to appear)*
Jonas Dann, Gustavo Alonso, Yannis Chronis, Maximilian Heer, Sven Hepkema, Ana Klimovic, Vasilis Mageirakos, Benjamin Ramhorst, Silvia R. Alcaraz, **Bowen Wu**, Yu Zhu

Terabyte-Scale Analytics in the Blink of an Eye *VLDB 2026*
Bowen Wu*, Wei Cui*, Carlo Curino, Matteo Interlandi, Rathijit Sen

Powerful GPUs or Fast Interconnects: Analyzing Relational Workloads on Modern GPUs *VLDB 2025*
Marko Kabić, **Bowen Wu**, Jonas Dann, Gustavo Alonso

Cracking Vector Search Indexes *VLDB 2025*
Vasilis Mageirakos, **Bowen Wu**, Gustavo Alonso

Efficiently Processing Joins and Grouped Aggregations on GPUs *SIGMOD 2025*
Bowen Wu, Dimitrios Koutsoukos, Gustavo Alonso

An Information-Theoretic Measure for Enabling Category Exemptions with an Application to Filter Bubbles *BIAS @ ECIR 2021*
Bowen Wu*, Chenyu Jiang*, Sanghamitra Dutta, Pulkit Grover

High Performance Distributed OLAP over Property Graphs with Grasper *SIGMOD 2020 (Demo)*
Hongzhi Chen, **Bowen Wu**, Shiyuan Deng, Chenghuan Huang, Changji Li, Chaoyi Li, James Cheng.

Under Submission

Over the Memory Wall, Into the Instruction Wall: The New Bottleneck in GPU Data Processing *arXiv:2608.13696*
Sven Hepkema, **Bowen Wu**, Christos Kozyrakis, Yannis Chronis, Gustavo Alonso

Eiger: An Efficient Library for GPU-based Data Analytics *arXiv:2607.04489*
Bowen Wu, Marko Kabić, Sven Hepkema, Vasilis Mageirakos, Gustavo Alonso

To GPU or Not to GPU: Vector Search in Relational Engines *arXiv:2605.15957*
Vasilis Mageirakos, Joel André, Marko Kabić, **Bowen Wu**, Yannis Chronis, Gustavo Alonso

*Equal contribution.

Invited Talks

NVIDIA cuDF Team	Sep 2026	**Microsoft Gray Systems Lab**	Aug 2026
Meta Velox Team	Sep 2025	**Huawei Edinburgh Research Center**	Mar 2025
IBM Research Center	Mar 2025	**Microsoft Gray Systems Lab**	Jun 2024

Teaching

Data Management Systems, *Teaching Assistant & Invited Speaker* Autumn 2024, 2025, 2026
Data Modeling and Databases, *Teaching Assistant* Spring 2024, 2025, 2026
Hardware Acceleration for Data Processing (Seminar), *Invited Speaker* Autumn 2024
Information Systems for Engineers, *Teaching Assistant* Autumn 2023

Selected Projects

Eiger: GPU Relational Operator Library
Lead developer, primary PhD project

- Built a GPU-accelerated relational query engine with highly optimized database operators, covering all 22 TPC-H queries and supporting multi-GPU execution. [GitHub]
- Proposed runtime adaptivity that performs adaptive algorithm selection and on-the-fly data transformation.
- Achieved up to 6.1× speedup over NVIDIA cuDF.

SKEIN: Semantic SQL Queries with Local LLMs
Project Co-lead

- Developing a DuckDB community extension that adds LLM-powered semantic operators to SQL, running on locally deployed models using vLLM. [GitHub]
- Achieved up to 8.2× higher throughput than the Flock DuckDB extension with lower resource usage and uncompromised accuracy.

Technical Skills

Languages	C/C++, CUDA (also PTX and SASS), Python, SQL, Shell
Systems	DuckDB, vLLM, SGLang
Libraries	MPI, NCCL, CCCL, CUB, cuCollections, cuDF
Tools	Nsight Systems, Nsight Compute
Topics	GPU Programming, Databases, Distributed Systems, Systems for ML, Computer Architecture